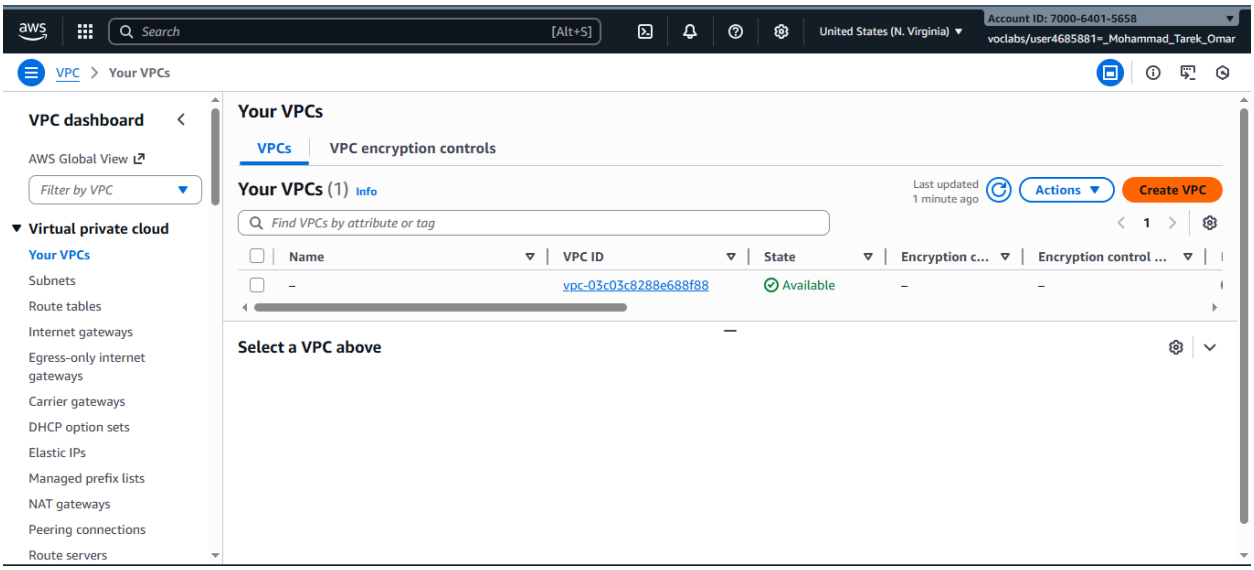
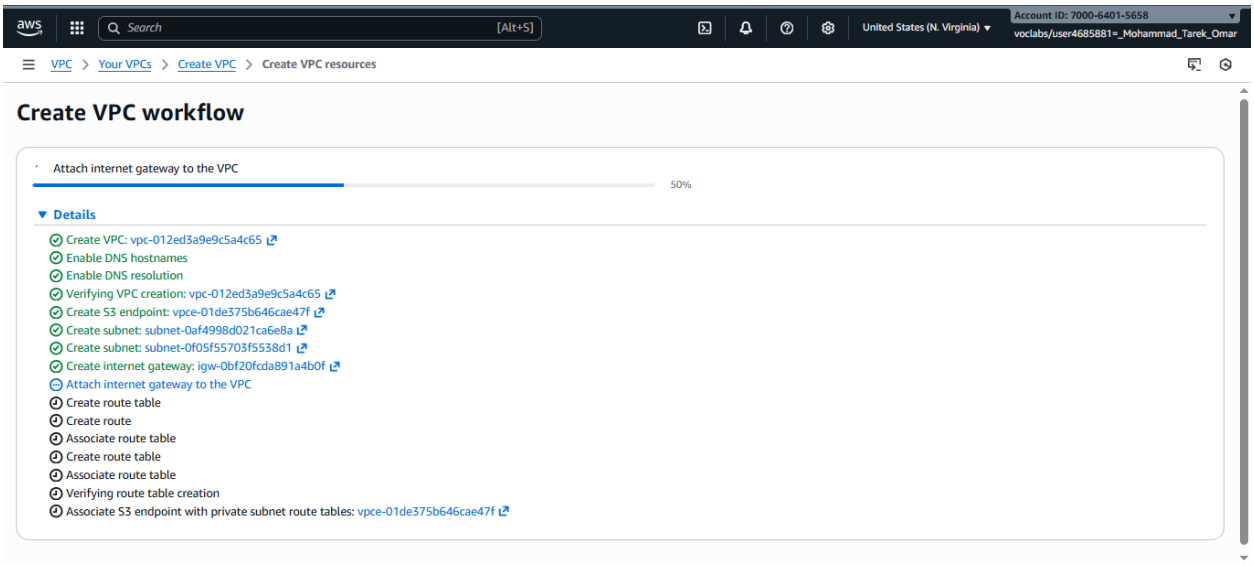
ITI – Cloud Lab 1

Mohammad Tarek Omar – Data Management





aws | Search [Alt+S] | United States (N. Virginia) | Account ID: 7000-6401-5658 | voclabs/user4685881=_Mohammad_Tarek_Omar

VPC > Your VPCs > Create VPC > Create VPC resources

Create VPC workflow

Success

▼ Details

- Create VPC: vpc-012ed3a9e9c5a4c65
- Enable DNS hostnames
- Enable DNS resolution
- Verifying VPC creation: vpc-012ed3a9e9c5a4c65
- Create S3 endpoint: vpce-01de375b646cae47f
- Create subnet: subnet-0af4998d021ca6e8a
- Create subnet: subnet-0f05f55703f5538d1
- Create internet gateway: igw-0bf20fcda891a4b0f
- Attach internet gateway to the VPC
- Create route table: rtb-0679e2ed4fd1d73cb
- Create route
- Associate route table
- Create route table: rtb-0e4e105912743f31c
- Associate route table
- Verifying route table creation
- Associate S3 endpoint with private subnet route tables: vpce-01de375b646cae47f

Related VPC resources and services

vpc-012ed3a9e9c5a4c65 / ITI-Cloud-Lab1-vpc

Actions

Details Info

VPC ID vpc-012ed3a9e9c5a4c65	**State** Available	**Block Public Access** Off	**DNS hostnames** Enabled
DNS resolution Enabled	**Tenancy** default	**DHCP option set** dopt-02fa10db92a6c58e9	**Main route table** rtb-09b145cd3e2b2d35d
Main network ACL acl-0109cce2007e46a10	**Default VPC** No	**IPv4 CIDR** 10.0.0.0/16	**IPv6 pool** –
IPv6 CIDR (Network border group) –	**Network Address Usage metrics** Disabled	**Route 53 Resolver DNS Firewall rule groups** Failed to load rule groups	**Owner ID** 700064015658
Encryption control ID –	**Encryption control mode** –		

Resource map | CIDRs | Flow logs | Tags | Integrations

Resource map Info

Show all details

VPC
Your AWS virtual network
- ITI-Cloud-Lab1-vpc

Subnets (2)
Subnets within this VPC
us-east-1a
- ITI-Cloud-Lab1-subnet-public1-us-east-1a
- ITI-Cloud-Lab1-subnet-private1-us-east-1a

Route tables (3)
Route network traffic to resources
- rtb-09b145cd3e2b2d35d
- ITI-Cloud-Lab1-rtb-public
- ITI-Cloud-Lab1-rtb-private1-us-east-1a

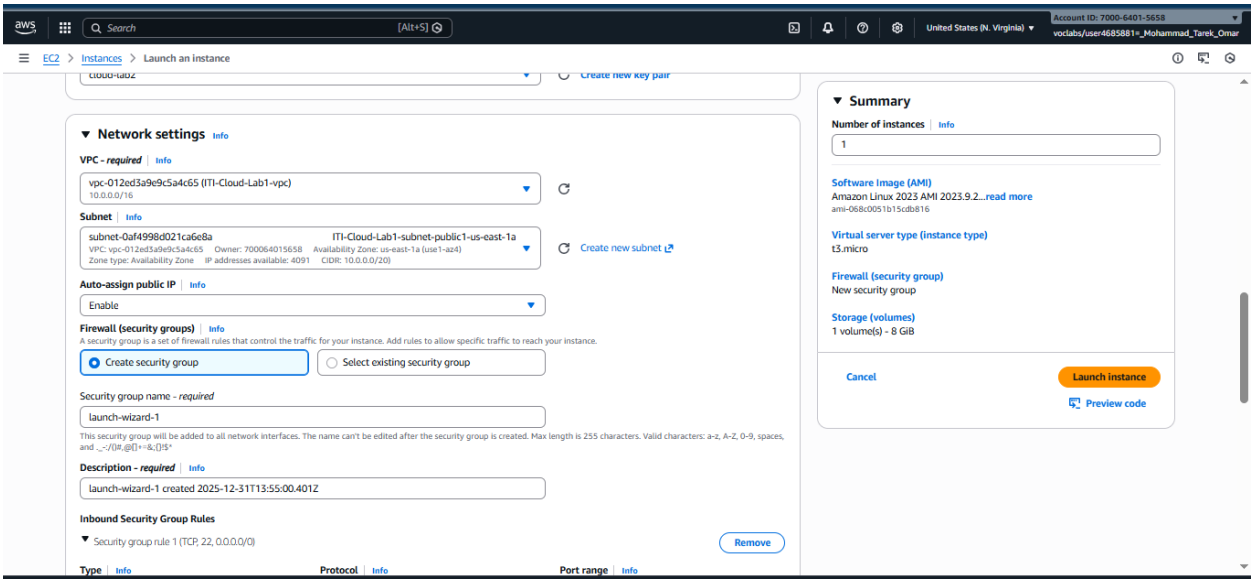
Network Connections (2)
Connections to other networks
- ITI-Cloud-Lab1-igw
- ITI-Cloud-Lab1-vpce-s3

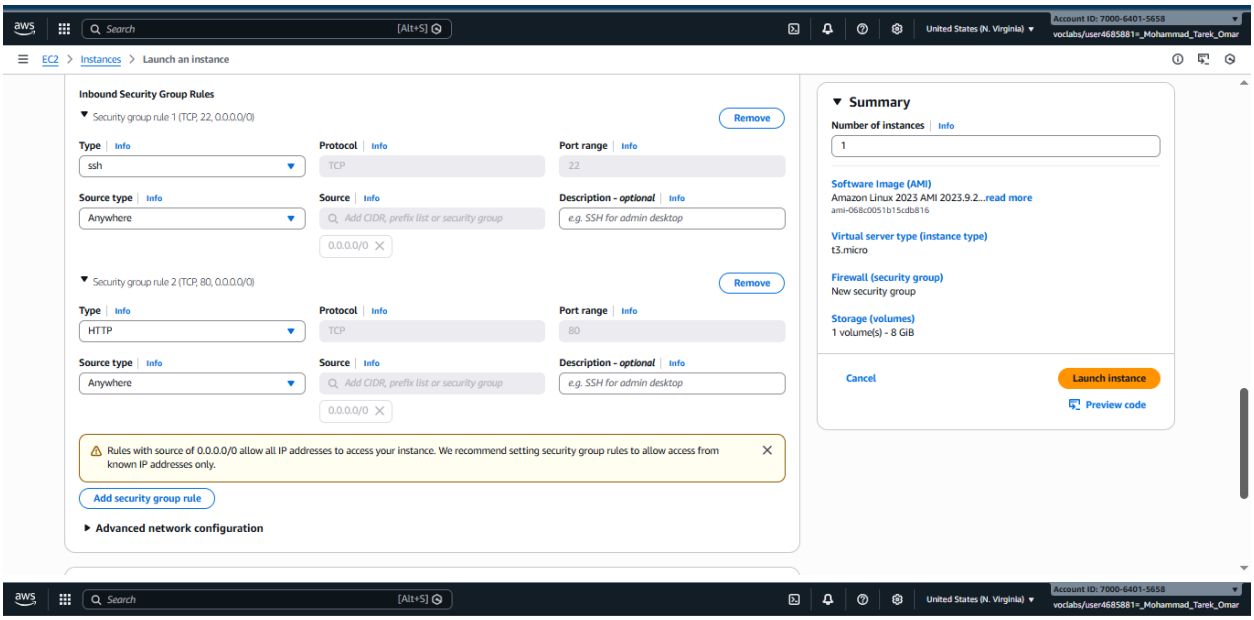
Subnets (8) Info

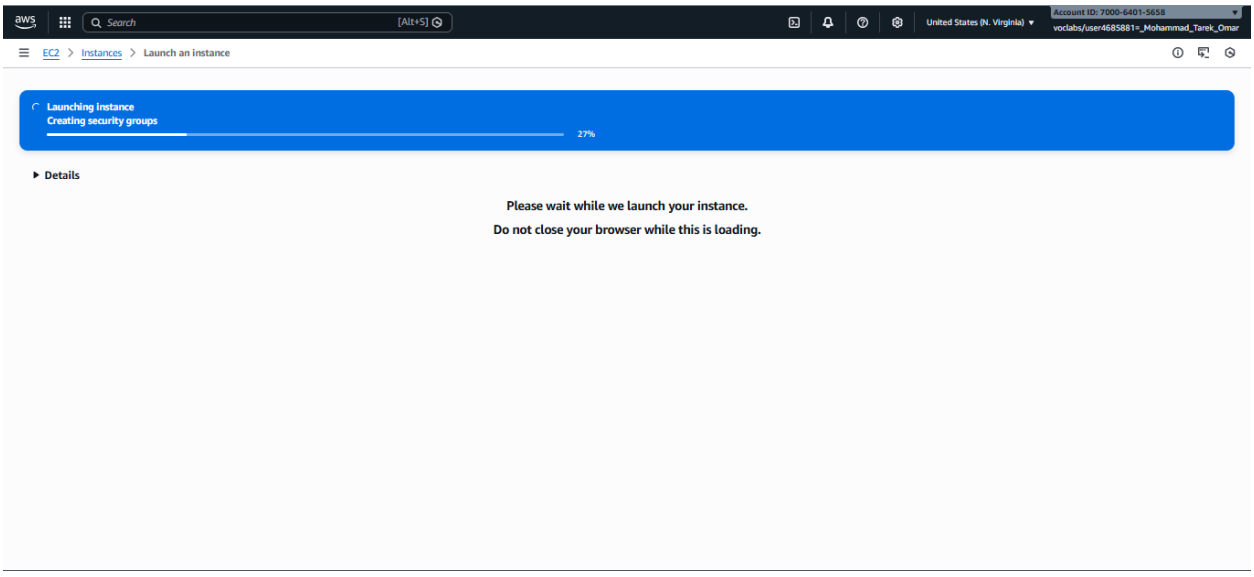
Last updated less than a minute ago | Actions | Create subnet

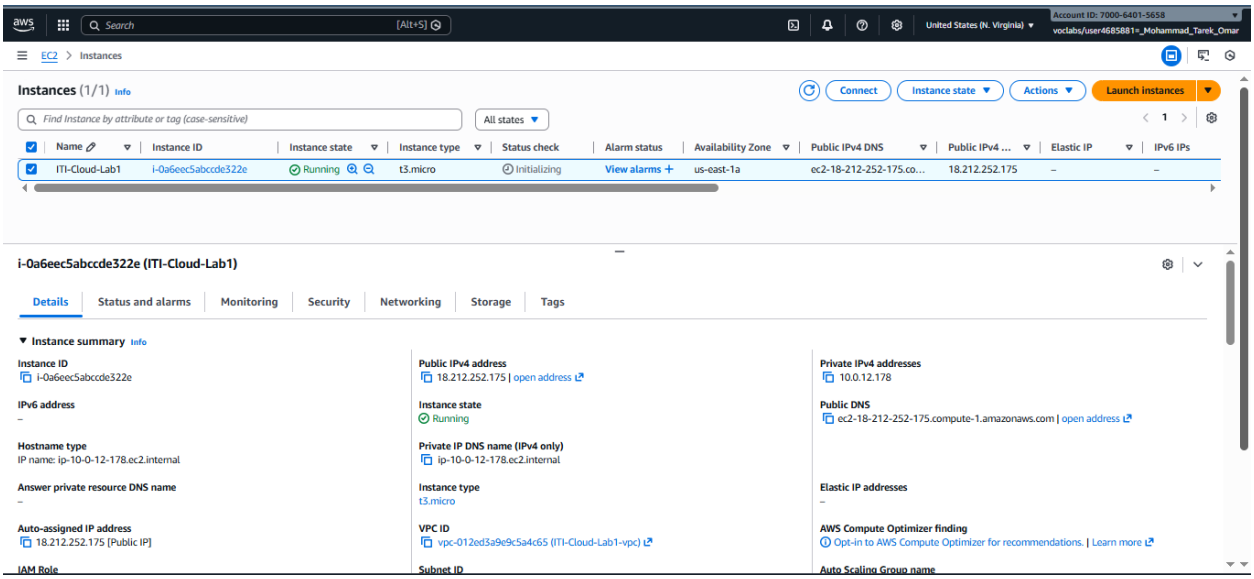
Find subnets by attribute or tag

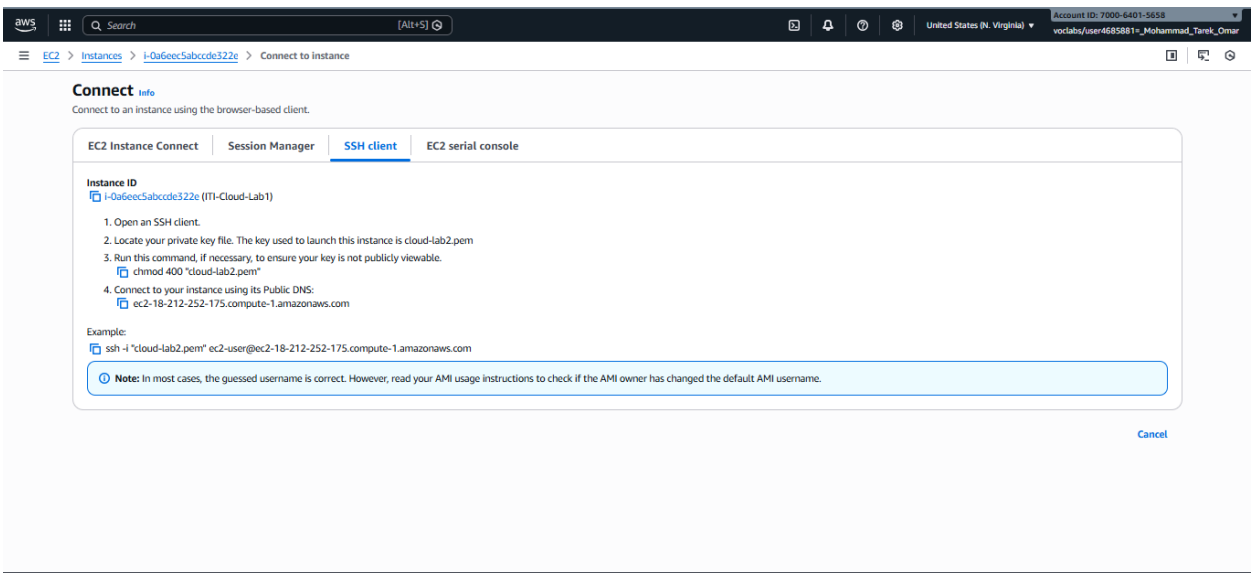
Name	Subnet ID	State	VPC	Block Public...	IPv4 CIDR	IPv6 CIDR
–	subnet-0ce1175a8a85ff9b1	Available	vpc-03c03c8288e688f88	Off	172.31.48.0/20	–
–	subnet-0fe6cc4cd49b42189	Available	vpc-03c03c8288e688f88	Off	172.31.80.0/20	–
–	subnet-082e5d7b01a615a71	Available	vpc-03c03c8288e688f88	Off	172.31.32.0/20	–
ITI-Cloud-Lab1-subnet-private1-us-east...	subnet-0f05f55703f5538d1	Available	vpc-012ed3a9e9c5a4c65 | ITI-C...	Off	10.0.128.0/20	–
–	subnet-08ab3e75c4667a3e0	Available	vpc-03c03c8288e688f88	Off	172.31.64.0/20	–
ITI-Cloud-Lab1-subnet-public1-us-east-1a	subnet-0af4998d021ca6e8a	Available	vpc-012ed3a9e9c5a4c65 | ITI-C...	Off	10.0.0.0/20	–
–	subnet-006c7a353403eaeca	Available	vpc-03c03c8288e688f88	Off	172.31.0.0/20	–
–	subnet-05dfc4b2acb1664db	Available	vpc-03c03c8288e688f88	Off	172.31.16.0/20	–

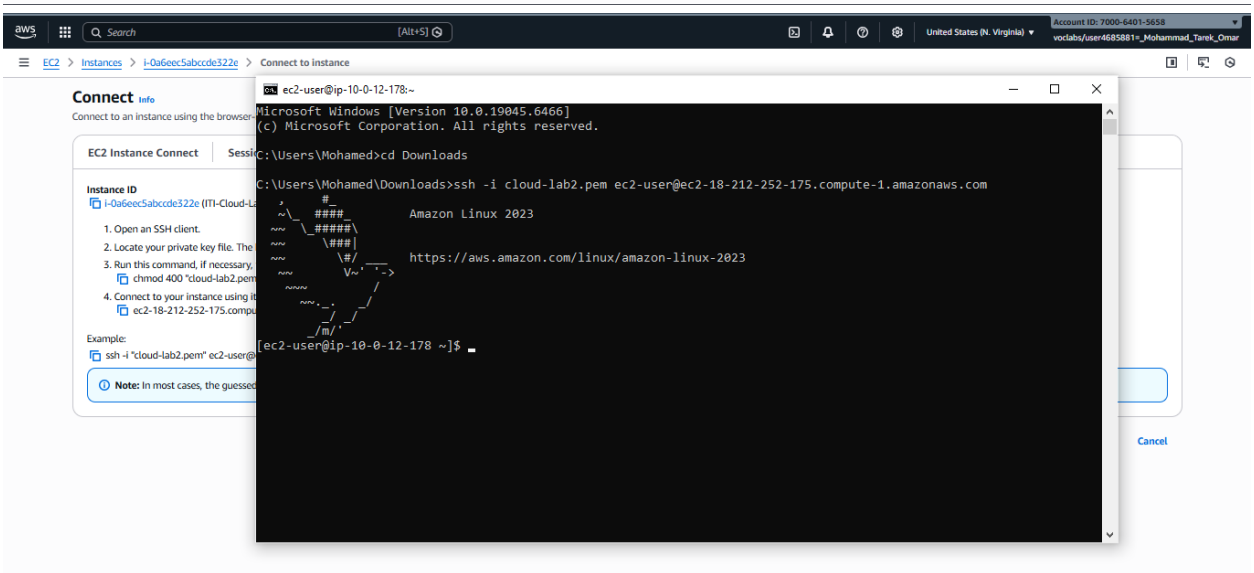

aws
Search
[Alt+S]
United States (N. Virginia)
Account ID: 7000-6401-5658
voclabs/user4685881=_Mohammad_Tarek_Omar
EC2 > Instances > Launch an instance
Create new key pair
Network settings Info
VPC - required Info
vpc-012ed3a9e9c5a4c65 (ITI-Cloud-Lab1-vpc)
10.0.0.0/16
Subnet Info
subnet-0af4998d021ca6e8a ITI-Cloud-Lab1-subnet-public1-us-east-1a
VPC: vpc-012ed3a9e9c5a4c65 Owner: 700064015658 Availability Zone: us-east-1a (use1-az4)
Zone type: Availability Zone IP addresses available: 4091 CIDR: 10.0.0.0/20)
Create new subnet
Auto-assign public IP Info
Enable
Firewall (security groups) Info
A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.
Create security group
Select existing security group
Security group name - required
launch-wizard-1
This security group will be added to all network interfaces. The name can't be edited after the security group is created. Max length is 255 characters. Valid characters: a-z, A-Z, 0-9, spaces, and ._-:/()#,@[]+=&;{}!$*
Description - required Info
launch-wizard-1 created 2025-12-31T13:55:00.401Z
Inbound Security Group Rules
Security group rule 1 (TCP, 22, 0.0.0.0/0)
Remove
Type Info
Protocol Info
Port range Info
Summary
Number of instances Info
1
Software Image (AMI)
Amazon Linux 2023 AMI 2023.9.2...read more
ami-068c0051b15cdb816
Virtual server type (instance type)
t3.micro
Firewall (security group)
New security group
Storage (volumes)
1 volume(s) - 8 GiB
Cancel
Launch instance
Preview code


aws
Search
[Alt+S]
United States (N. Virginia)
Account ID: 7000-6401-5658
voclabs/user4685881=_Mohammad_Tarek_Omar
EC2 > Instances > Launch an instance
Inbound Security Group Rules
Security group rule 1 (TCP, 22, 0.0.0.0/0)
Remove
Type Info
ssh
Protocol Info
TCP
Port range Info
22
Source type Info
Anywhere
Source Info
Add CIDR, prefix list or security group
0.0.0.0/0
Description - optional Info
e.g. SSH for admin desktop
Security group rule 2 (TCP, 80, 0.0.0.0/0)
Remove
Type Info
HTTP
Protocol Info
TCP
Port range Info
80
Source type Info
Anywhere
Source Info
Add CIDR, prefix list or security group
0.0.0.0/0
Description - optional Info
e.g. SSH for admin desktop
Rules with source of 0.0.0.0/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.
Add security group rule
Advanced network configuration
Summary
Number of instances Info
1
Software Image (AMI)
Amazon Linux 2023 AMI 2023.9.2...read more
ami-068c0051b15cdb816
Virtual server type (instance type)
t3.micro
Firewall (security group)
New security group
Storage (volumes)
1 volume(s) - 8 GiB
Cancel
Launch instance
Preview code


aws
Search
[Alt+S]
United States (N. Virginia)
Account ID: 7000-6401-5658
voclabs/user4685881=_Mohammad_Tarek_Omar
EC2 > Instances > Launch an instance
Launching instance
Creating security groups
27%
Details
Please wait while we launch your instance.
Do not close your browser while this is loading.


EC2 > Instances
Instances (1/1) Info
Connect
Instance state
Actions
Launch instances
Find Instance by attribute or tag (case-sensitive)
All states
Name
Instance ID
Instance state
Instance type
Status check
Alarm status
Availability Zone
Public IPv4 DNS
Public IPv4 ...
Elastic IP
IPv6 IPs
ITI-Cloud-Lab1
i-0a6eec5abccde322e
Running
t3.micro
Initializing
View alarms +
us-east-1a
ec2-18-212-252-175.co...
18.212.252.175
i-0a6eec5abccde322e (ITI-Cloud-Lab1)
Details
Status and alarms
Monitoring
Security
Networking
Storage
Tags
Instance summary Info
Instance ID
i-0a6eec5abccde322e
IPv6 address
Hostname type
IP name: ip-10-0-12-178.ec2.internal
Answer private resource DNS name
Auto-assigned IP address
18.212.252.175 [Public IP]
IAM Role
Public IPv4 address
18.212.252.175 | open address
Instance state
Running
Private IP DNS name (IPv4 only)
ip-10-0-12-178.ec2.internal
Instance type
t3.micro
VPC ID
vpc-012ed3a9e9c5a4c65 (ITI-Cloud-Lab1-vpc)
Subnet ID
Private IPv4 addresses
10.0.12.178
Public DNS
ec2-18-212-252-175.compute-1.amazonaws.com | open address
Elastic IP addresses
AWS Compute Optimizer finding
Opt-in to AWS Compute Optimizer for recommendations. | Learn more
Auto Scaling Group name


EC2 > Instances > i-0a6eec5abccde322e > Connect to instance
Connect Info
Connect to an instance using the browser-based client.
EC2 Instance Connect
Session Manager
SSH client
EC2 serial console
Instance ID
i-0a6eec5abccde322e (ITI-Cloud-Lab1)
1. Open an SSH client.
2. Locate your private key file. The key used to launch this instance is cloud-lab2.pem
3. Run this command, if necessary, to ensure your key is not publicly viewable.
chmod 400 "cloud-lab2.pem"
4. Connect to your instance using its Public DNS:
ec2-18-212-252-175.compute-1.amazonaws.com
Example:
ssh -i "cloud-lab2.pem" ec2-user@ec2-18-212-252-175.compute-1.amazonaws.com
Note: In most cases, the guessed username is correct. However, read your AMI usage instructions to check if the AMI owner has changed the default AMI username.
Cancel


EC2 > Instances > i-0a6eec5abccde322e > Connect to instance
Connect Info
ec2-user@ip-10-0-12-178:~
Microsoft Windows [Version 10.0.19045.6466]
(c) Microsoft Corporation. All rights reserved.
C:\Users\Mohamed>cd Downloads
C:\Users\Mohamed\Downloads>ssh -i cloud-lab2.pem ec2-user@ec2-18-212-252-175.compute-1.amazonaws.com
Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023
[ec2-user@ip-10-0-12-178 ~]$
Cancel

ec2-user@ip-10-0-12-178:~

```
Microsoft Windows [Version 10.0.19045.6466]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Mohamed>cd Downloads

C:\Users\Mohamed\Downloads>ssh -i cloud-lab2.pem ec2-user@ec2-18-212-252-175.compute-1.amazonaws.com
       ,     #_
       ~\_  ####_        Amazon Linux 2023
      ~~  \_#####\
      ~~     \###|
      ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
       ~~       V~' '->
        ~~~         /
          ~~._.   _/
             _/ _/
           _/m/'
[ec2-user@ip-10-0-12-178 ~]$ touch script
[ec2-user@ip-10-0-12-178 ~]$ chmod u+x script
[ec2-user@ip-10-0-12-178 ~]$ touch script.sh
[ec2-user@ip-10-0-12-178 ~]$ chmod u+x script.sh
[ec2-user@ip-10-0-12-178 ~]$ vim script.sh
```

ec2-user@ip-10-0-12-178:~

```
Microsoft Windows [Version 10.0.19045.6466]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Mohamed>cd Downloads

C:\Users\Mohamed\Downloads>ssh -i cloud-lab2.pem ec2-user@ec2-18-212-252-175.compute-1.amazonaws.com
       ,     #_
       ~\_  ####_        Amazon Linux 2023
      ~~  \_#####\
      ~~     \###|
      ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
       ~~       V~' '->
        ~~~         /
          ~~._.   _/
             _/ _/
           _/m/'
[ec2-user@ip-10-0-12-178 ~]$ touch script
[ec2-user@ip-10-0-12-178 ~]$ chmod u+x script
[ec2-user@ip-10-0-12-178 ~]$ touch script.sh
[ec2-user@ip-10-0-12-178 ~]$ chmod u+x script.sh
[ec2-user@ip-10-0-12-178 ~]$ vim script.sh
[ec2-user@ip-10-0-12-178 ~]$ [ec2-user@ip-10-0-12-178 ~]$ sudo ./script.sh
```

ec2-user@ip-10-0-12-178:~

```
  Installing       : generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch                    12/13
  Installing       : httpd-2.4.65-1.amzn2023.0.2.x86_64                                   13/13
  Running scriptlet: httpd-2.4.65-1.amzn2023.0.2.x86_64                                   13/13
  Verifying        : apr-1.7.5-1.amzn2023.0.4.x86_64                                       1/13
  Verifying        : apr-util-1.6.3-1.amzn2023.0.2.x86_64                                  2/13
  Verifying        : apr-util-lmdb-1.6.3-1.amzn2023.0.2.x86_64                             3/13
  Verifying        : apr-util-openssl-1.6.3-1.amzn2023.0.2.x86_64                          4/13
  Verifying        : generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch                     5/13
  Verifying        : httpd-2.4.65-1.amzn2023.0.2.x86_64                                    6/13
  Verifying        : httpd-core-2.4.65-1.amzn2023.0.2.x86_64                               7/13
  Verifying        : httpd-filesystem-2.4.65-1.amzn2023.0.2.noarch                         8/13
  Verifying        : httpd-tools-2.4.65-1.amzn2023.0.2.x86_64                              9/13
  Verifying        : libbrotli-1.0.9-4.amzn2023.0.2.x86_64                                10/13
  Verifying        : mailcap-2.1.49-3.amzn2023.0.3.noarch                                 11/13
  Verifying        : mod_http2-2.0.27-1.amzn2023.0.3.x86_64                               12/13
  Verifying        : mod_lua-2.4.65-1.amzn2023.0.2.x86_64                                 13/13

Installed:
  apr-1.7.5-1.amzn2023.0.4.x86_64                      apr-util-1.6.3-1.amzn2023.0.2.x86_64
  apr-util-lmdb-1.6.3-1.amzn2023.0.2.x86_64            apr-util-openssl-1.6.3-1.amzn2023.0.2.x86_64
  generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch    httpd-2.4.65-1.amzn2023.0.2.x86_64
  httpd-core-2.4.65-1.amzn2023.0.2.x86_64              httpd-filesystem-2.4.65-1.amzn2023.0.2.noarch
  httpd-tools-2.4.65-1.amzn2023.0.2.x86_64             libbrotli-1.0.9-4.amzn2023.0.2.x86_64
  mailcap-2.1.49-3.amzn2023.0.3.noarch                 mod_http2-2.0.27-1.amzn2023.0.3.x86_64
  mod_lua-2.4.65-1.amzn2023.0.2.x86_64

Complete!
Created symlink /etc/systemd/system/multi-user.target.wants/httpd.service → /usr/lib/systemd/system/httpd
.service.
[ec2-user@ip-10-0-12-178 ~]$
```

Not secure 18.212.252.175

ITI English DE Roadmap YouTube translate ChatGPT ITI DM Slides

This is Mohammad Tarek from DM ITI Cloud course